

14.2 Best linear prediction

With random rows, a regression is a rule for predicting the response of a new case from its regressors. The population theory is in place: the conditional mean $m(\mathbf{X}) = E(Y | \mathbf{X})$ is the best predictor (Proposition 6.11.2), the best *linear* predictor $L(Y | \mathbf{X}) = \alpha^* + \boldsymbol{\beta}^{*\top} \mathbf{X}$ is a projection (Theorem 6.11.1, Theorem 2.5.2), the two coincide for jointly normal variables (Proposition 3.5.3), and least squares estimates L consistently (Proposition 6.11.3). This section asks the finite-sample questions: how large is the error of the estimated predictor on a new case, and how misleading is its error on the data used to fit it?

14.2.1 Characterizing the best predictor

The next result gives a test for whether a predictor is the best one, and a correlation that measures how well the best predictor can do.

Proposition 14.2.1 — The best predictor and the correlation ratio

Let $EY^2 < \infty$, $\text{Var}(Y) > 0$, and $m(\mathbf{X}) = E(Y | \mathbf{X})$.

- (a) A predictor $g(\mathbf{X})$ with $Eg(\mathbf{X})^2 < \infty$ equals $m(\mathbf{X})$ with probability one iff $Eg(\mathbf{X}) = EY$ and $\text{Cov}(h(\mathbf{X}), Y - g(\mathbf{X})) = 0$ for every square-integrable h .
- (b) For every such g with $\text{Var } g(\mathbf{X}) > 0$,

$$\text{corr}(Y, g(\mathbf{X}))^2 \leq \eta^2 = \frac{\text{Var } m(\mathbf{X})}{\text{Var } Y}, \quad (14.2.1)$$

with equality for $g = m$ when $\text{Var } m(\mathbf{X}) > 0$. The **correlation ratio** η^2 exceeds the squared multiple correlation $\rho_{Y \cdot X}^2$ of Proposition 2.5.3 by $E(m(\mathbf{X}) - L(Y | \mathbf{X}))^2 / \text{Var } Y$, so $\eta^2 = \rho_{Y \cdot X}^2$ iff m is affine.

Proof.

- (a) If $g = m$, then $Em(\mathbf{X}) = EY$ by the tower rule, and $E[(Y - m)h] = 0$ by Proposition 6.11.2(a), so the covariance is zero. Conversely, let $d = g - m$. Then $Ed = 0$ and, taking $h = d$, $\text{Var } d = \text{Cov}(d, Y - m) - \text{Cov}(d, Y - g) = 0 - 0$, so $d = 0$ with probability one.
- (b) By (a) with $h = g$, $\text{Cov}(Y, g) = \text{Cov}(m, g)$. Cauchy–Schwarz for covariances gives $\text{Cov}(m, g)^2 \leq \text{Var } m \text{ Var } g$, hence $\text{corr}(Y, g)^2 = \text{Cov}(m, g)^2 / (\text{Var } Y \text{ Var } g) \leq \text{Var } m / \text{Var } Y$. For $g = m$ the covariance is $\text{Var } m$ and equality holds. For the last claim write $L = L(Y | \mathbf{X}) = L(m | \mathbf{X})$ (Proposition 6.11.2(b)). The error $m - L$ has mean zero and is uncorrelated with the affine function L , so $\text{Var } m = \text{Var } L + E(m - L)^2$. Divide by $\text{Var } Y$ and use $\text{Var } L / \text{Var } Y = \rho_{Y \cdot X}^2$ (Eq. 6.11.1).

■

Part (a) is the population basis of residual plots. If a fitted model reproduces the conditional mean, its residuals are uncorrelated with *every* function of the regressors, not only with the

regressors themselves. A residual plot that shows a trend against x_j^2 , against a product $x_j x_k$ or against the fitted values is evidence that the fit is not m . Sections 14.6 and 14.7 turn this into tests. Part (b) says that no transformation of the regressors predicts better, in the correlation sense, than the conditional mean. The gap $\eta^2 - \rho_{Y \cdot X}^2$ is the share of the variance of Y that a linear predictor leaves on the table.

14.2.2 The error of an estimated predictor

In practice α^* and β^* are replaced by least squares estimates from a training sample $(\mathbf{X}_i, Y_i)$, $i = 1, \dots, n$. By centring and Theorem 6.6.2 these are the plug-in estimates $\hat{\beta} = \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}$ and $\hat{\alpha} = \bar{Y} - \hat{\beta}^\top \bar{\mathbf{X}}$, built from the sample means and covariances. Their performance is judged on a new case $(\mathbf{X}_0, Y_0)$ drawn from the same distribution, independently of the training data.

Proposition 14.2.2 — Excess error of a fitted linear predictor

Let $(\mathbf{X}_0, Y_0)$ have finite second moments, $\text{Cov}(\mathbf{X}_0) = \Sigma_{XX}$, mean μ_X , and best linear predictor $L_0 = \alpha^* + \beta^{*\top} \mathbf{X}_0$ with error variance $\sigma_*^2 = \text{E}(Y_0 - L_0)^2$. Let $\hat{Y}_0 = a + \mathbf{b}^\top \mathbf{X}_0$, where $(a, \mathbf{b})$ depend only on training data independent of $(\mathbf{X}_0, Y_0)$. Then, given the training data,

$$\begin{aligned} \text{E}[(Y_0 - \hat{Y}_0)^2 \mid \text{training data}] &= \sigma_*^2 + (\alpha^* - a + (\beta^* - \mathbf{b})^\top \mu_X)^2 \\ &\quad + (\beta^* - \mathbf{b})^\top \Sigma_{XX} (\beta^* - \mathbf{b}). \end{aligned}$$

Proof. Condition on the training data, so that a and $\mathbf{b}$ are constants. Write $Y_0 - \hat{Y}_0 = (Y_0 - L_0) + (L_0 - \hat{Y}_0)$. The second term is an affine function of $\mathbf{X}_0$, and by Theorem 6.11.1 the first has mean zero and is uncorrelated with every affine function of $\mathbf{X}_0$. So the cross term vanishes, and the mean square of the affine term is its squared mean plus its variance $(\beta^* - \mathbf{b})^\top \Sigma_{XX} (\beta^* - \mathbf{b})$. ■

The first term is the error of the best linear predictor, which no amount of data can remove. The other two are the price of estimation, and by Proposition 6.11.3 they vanish as $n \rightarrow \infty$ whatever the joint distribution. To see how large they are for moderate n , we need the distribution of the regressors. Under joint normality the calculation can be done exactly. It uses one new distribution.

Definition 14.2.1 — Wishart distribution

Let the rows of the $m \times k$ matrix $\mathbf{Z}$ be independent $N_k(\mathbf{0}, \Sigma)$ vectors. The law of $\mathbf{W} = \mathbf{Z}^\top \mathbf{Z} = \sum_{i=1}^m \mathbf{z}_i \mathbf{z}_i^\top$ is the **Wishart distribution** $W_k(m, \Sigma)$ with m degrees of freedom. For $k = 1$, $W_1(m, \sigma^2)$ is the law of $\sigma^2 \chi^2(m)$.

Lemma 14.2.3 — Centred cross-products

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ ($n \geq 2$) be independent $N_k(\mu, \Sigma)$ vectors with mean $\bar{\mathbf{X}}$, and let $\mathbf{W} =$

$\sum_i (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top$. Then $\mathbf{W} \sim W_k(n-1, \Sigma)$, and $\mathbf{W}$ is independent of $\bar{\mathbf{X}}$.

Proof. Let $\mathbf{X}_1$ be the $n \times k$ matrix with rows $\mathbf{X}_i^\top$. Write $\mathbf{X}_1 = \mathbf{1}\mu^\top + \mathbf{Z}_0\Sigma^{1/2}$, where $\Sigma^{1/2}$ is the symmetric square root (Theorem 1.7.6) and $\mathbf{Z}_0$ has independent standard normal entries. Let $\mathbf{O}$ be an orthogonal $n \times n$ matrix with first row $n^{-1/2}\mathbf{1}^\top$, for instance the Helmert matrix of Example (Helmert matrices), and let $\mathbf{K}$ consist of its last $n-1$ rows. From $\mathbf{O}^\top\mathbf{O} = \mathbf{I}$, $\mathbf{K}^\top\mathbf{K} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$, so $\mathbf{W} = \mathbf{X}_1^\top\mathbf{K}^\top\mathbf{K}\mathbf{X}_1 = (\mathbf{K}\mathbf{X}_1)^\top(\mathbf{K}\mathbf{X}_1)$, and $\mathbf{K}\mathbf{1} = \mathbf{0}$ gives $\mathbf{K}\mathbf{X}_1 = (\mathbf{K}\mathbf{Z}_0)\Sigma^{1/2}$.

Each column of $\mathbf{O}\mathbf{Z}_0$ is $N_n(\mathbf{0}, \mathbf{O}\mathbf{O}^\top) = N_n(\mathbf{0}, \mathbf{I})$ by Theorem 3.3.1, and the columns are independent, so $\mathbf{O}\mathbf{Z}_0$ again has independent standard normal entries. Its first row is $\sqrt{n}(\bar{\mathbf{X}} - \mu)^\top\Sigma^{-1/2}$, and its other rows form $\mathbf{K}\mathbf{Z}_0$. Hence the $n-1$ rows of $\mathbf{K}\mathbf{X}_1$ are independent $N_k(\mathbf{0}, \Sigma)$ vectors, independent of $\bar{\mathbf{X}}$, and $\mathbf{W} \sim W_k(n-1, \Sigma)$. ■

Lemma 14.2.4 — Mean of an inverse Wishart matrix

Let $\mathbf{W} \sim W_k(m, \Sigma)$ with Σ positive definite and $m \geq k+2$. Then $\mathbf{W}$ is nonsingular with probability one, and

$$E(\mathbf{W}^{-1}) = \frac{\Sigma^{-1}}{m-k-1}.$$

Proof. Reduction. Write $\mathbf{Z} = \mathbf{Z}_0\Sigma^{1/2}$ with $\mathbf{Z}_0$ as in the previous proof. Then $\mathbf{W} = \Sigma^{1/2}\mathbf{W}_0\Sigma^{1/2}$ with $\mathbf{W}_0 = \mathbf{Z}_0^\top\mathbf{Z}_0 \sim W_k(m, \mathbf{I})$, and $\mathbf{W}^{-1} = \Sigma^{-1/2}\mathbf{W}_0^{-1}\Sigma^{-1/2}$. It suffices to show $E(\mathbf{W}_0^{-1}) = \mathbf{I}/(m-k-1)$.

Nonsingularity. $\mathbf{W}_0$ is singular iff some column $\mathbf{z}_j$ of $\mathbf{Z}_0$ lies in the span of the others. Given the other columns, that span has dimension at most $k-1 < m$, and $\mathbf{z}_j \sim N_m(\mathbf{0}, \mathbf{I})$, which has a density, falls in it with probability zero.

Diagonal entries. Partition $\mathbf{Z}_0 = [\mathbf{z}_1, \mathbf{Z}_2]$. By Theorem 1.6.4, the $(1,1)$ entry of $\mathbf{W}_0^{-1}$ is $1/(\mathbf{z}_1^\top\mathbf{z}_1 - \mathbf{z}_1^\top\mathbf{Z}_2(\mathbf{Z}_2^\top\mathbf{Z}_2)^{-1}\mathbf{Z}_2^\top\mathbf{z}_1) = 1/\mathbf{z}_1^\top(\mathbf{I} - \mathbf{P}_2)\mathbf{z}_1$, where $\mathbf{P}_2$ projects onto $\mathcal{C}(\mathbf{Z}_2)$, of dimension $k-1$. Given $\mathbf{Z}_2$, $\mathbf{z}_1$ is still $N_m(\mathbf{0}, \mathbf{I})$, so $\mathbf{z}_1^\top(\mathbf{I} - \mathbf{P}_2)\mathbf{z}_1 \sim \chi^2(m-k+1)$ by Theorem 4.3.2(a). This law does not depend on $\mathbf{Z}_2$, so it holds unconditionally (Lemma 14.1.1), and Lemma 14.1.3 gives mean $1/(m-k-1)$, finite because $m-k+1 > 2$. The same argument applies to every diagonal entry.

Off-diagonal entries. Let $\mathbf{D}_i$ be the identity with its i th diagonal entry replaced by -1 . Then $\mathbf{Z}_0\mathbf{D}_i$ has the same law as $\mathbf{Z}_0$, so $\mathbf{D}_i\mathbf{W}_0^{-1}\mathbf{D}_i = (\mathbf{D}_i\mathbf{W}_0\mathbf{D}_i)^{-1}$ has the same law as $\mathbf{W}_0^{-1}$. The mean of an off-diagonal entry (i,j) therefore equals its own negative, and is zero. It exists because $|w^{ij}| \leq \frac{1}{2}(w^{ii} + w^{jj})$ for the entries of the positive definite $\mathbf{W}_0^{-1}$. ■

With these two lemmas the random-design analogue of Theorem 5.5.1 follows at once.

Corollary 14.2.5 — Covariance of the slopes under normal regressors

Let the rows $(\mathbf{X}_i^\top, Y_i)$ be independent draws from a $(k+1)$ -variate normal distribution with Σ_{XX} positive definite and conditional variance $\sigma^2 = \text{Var}(Y \mid \mathbf{X})$, and let $\hat{\boldsymbol{\beta}}$ be the least squares slopes from a fit with an intercept. If $n \geq k+3$,

$$\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{n-k-2} \Sigma_{XX}^{-1}.$$

Proof. Given the regressors, the slopes are unbiased with covariance $\sigma^2 \mathbf{W}^{-1}$, where $\mathbf{W}$ is the centred cross-product matrix (Eq. 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$). By Lemma 14.2.3 $\mathbf{W} \sim W_k(n-1, \Sigma_{XX})$, and by Proposition 2.5.4 and Lemma 14.2.4 the unconditional covariance is $\sigma^2 \mathbb{E}(\mathbf{W}^{-1}) = \sigma^2 \Sigma_{XX}^{-1} / (n-k-2)$. ■

For $k=1$ this is Eq. 14.1.2. The factor $1/(n-k-2)$, against $1/(n-1)$ for a fixed design with the average spread, grows quickly with k .

Theorem 14.2.6 — Prediction error under normal regressors

In the setting of Corollary 14.2.5, let $(\mathbf{X}_0, Y_0)$ be a further independent draw and $\hat{Y}_0 = \hat{\alpha} + \hat{\boldsymbol{\beta}}^\top \mathbf{X}_0$ the fitted predictor. If $n \geq k+3$, then

$$\mathbb{E}(Y_0 - \hat{Y}_0)^2 = \sigma^2 \left(1 + \frac{1}{n}\right) \frac{n-2}{n-k-2}. \quad (14.2.2)$$

Proof. Write $\hat{Y}_0 = \bar{Y} + \hat{\boldsymbol{\beta}}^\top (\mathbf{X}_0 - \bar{\mathbf{X}})$, and use the model $Y_i = \alpha + \boldsymbol{\beta}^\top \mathbf{X}_i + e_i$ of Eq. 3.5.2 for $i = 0, \dots, n$. Since $\bar{Y} = \alpha + \boldsymbol{\beta}^\top \bar{\mathbf{X}} + \bar{e}$,

$$Y_0 - \hat{Y}_0 = e_0 - \bar{e} - (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top (\mathbf{X}_0 - \bar{\mathbf{X}}).$$

Condition on $\mathbf{X}_0, \dots, \mathbf{X}_n$. Then e_0 , $\bar{e}$ and $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \mathbf{W}^{-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}}) e_i$ are normal with mean zero, and they are uncorrelated: e_0 is independent of the training errors, and $\text{Cov}(\bar{e}, \hat{\boldsymbol{\beta}}) = \sigma^2 n^{-1} \mathbf{W}^{-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}}) = \mathbf{0}$. So the conditional mean square is

$$\sigma^2 \left[1 + \frac{1}{n} + (\mathbf{X}_0 - \bar{\mathbf{X}})^\top \mathbf{W}^{-1} (\mathbf{X}_0 - \bar{\mathbf{X}}) \right].$$

Now $\mathbf{u} = \mathbf{X}_0 - \bar{\mathbf{X}} \sim N_k(\mathbf{0}, (1 + 1/n) \Sigma_{XX})$, and $\mathbf{u}$ is independent of $\mathbf{W}$, because $\mathbf{X}_0$ is independent of the training sample and $\bar{\mathbf{X}}$ is independent of $\mathbf{W}$ (Lemma 14.2.3). Hence $\mathbb{E}(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}) = \text{tr}[\mathbb{E}(\mathbf{W}^{-1}) \mathbb{E}(\mathbf{u} \mathbf{u}^\top)] = (1 + 1/n) k / (n - k - 2)$ by Lemma 14.2.4, and

$$\mathbb{E}(Y_0 - \hat{Y}_0)^2 = \sigma^2 \left(1 + \frac{1}{n}\right) \left(1 + \frac{k}{n-k-2}\right) = \sigma^2 \left(1 + \frac{1}{n}\right) \frac{n-2}{n-k-2}. \quad \blacksquare$$

The conditional mean square in the proof, $\sigma^2 [1 + 1/n + (\mathbf{X}_0 - \bar{\mathbf{X}})^\top \mathbf{W}^{-1} (\mathbf{X}_0 - \bar{\mathbf{X}})]$, is the variance behind the prediction interval of Theorem 12.3.1, so that interval is exact for random regressors as well (Theorem 14.1.2). What the theorem adds is the average over new cases and over training designs.

14.2.3 The optimism of the in-sample error

The residual mean square SSE/n is the error the fitted predictor makes on the cases used to fit it. It is a biased guide to the error on new cases, in both the fixed and the random design.

Proposition 14.2.7 — In-sample and out-of-sample error

- (a) (*Fixed design.*) Let $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$ with $\text{rank}(\mathbf{X}) = r$, and let $\mathbf{Y}^*$ be a vector of new responses at the same design, uncorrelated with $\mathbf{Y}$, with the same mean and covariance. Then

$$E \frac{\|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2}{n} = \sigma^2 \frac{n-r}{n}, \quad E \frac{\|\mathbf{Y}^* - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2}{n} = \sigma^2 \frac{n+r}{n},$$

so the in-sample error understates the error on new responses by $2r\sigma^2/n$ on average.

- (b) (*Normal random design.*) In the setting of [Theorem 14.2.6](#), the in-sample error has $E(\text{SSE}/n) = \sigma^2(n-k-1)/n$, and the error on a new case is [Eq. 14.2.2](#).

Proof.

- (a) The first identity is $E(\text{SSE}) = \sigma^2(n-r)$ (Theorem 5.6.1, or Eq. 2.3.1 with a mean in $\mathcal{C}(\mathbf{X})$). For the second, $\mathbf{Y}^* - \mathbf{X}\hat{\boldsymbol{\beta}} = (\mathbf{Y}^* - \mathbf{X}\boldsymbol{\beta}) - \mathbf{M}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$, and the two terms are uncorrelated, so the expected squared length is $n\sigma^2 + \sigma^2 \text{tr}(\mathbf{M}) = \sigma^2(n+r)$ by Theorem 2.3.1.
- (b) Given the regressors, $E(\text{SSE} \mid \mathbf{X}) = \sigma^2(n-k-1)$, which does not depend on $\mathbf{X}$; take expectations. ■

A new random case brings its own, possibly outlying, regressor values, where the fitted plane is least reliable, so the random design is worse. [Figure 14.2.1](#) shows the three expected errors for $n = 30$. With $k = 10$ regressors the in-sample error is $0.633\sigma^2$, the error on new responses at the same design is $1.367\sigma^2$, and the error on a new random case is $1.607\sigma^2$: the in-sample figure flatters the fit by a factor of about 2.5. At $k = 20$ the three values are 0.300, 1.700 and 3.617, in units of σ^2 . The simulated averages over 20000 training samples per point, for instance 1.604 at $k = 10$ and 3.612 at $k = 20$, agree with [Eq. 14.2.2](#). Estimating the out-of-sample error honestly, by penalties such as Mallows' C_p or by cross-validation, is the subject of [Chapter 29](#).

```
import numpy as np
rng = np.random.default_rng(1422)
n, sigma2, reps = 30, 1.0, 20_000

def simulate(k):
    """Average in-sample and out-of-sample squared error over many training samples.
    """
    ins, out = np.empty(reps), np.empty(reps)
    beta = np.full(k, 0.5)
    for s in range(reps):
        X = np.column_stack([np.ones(n), rng.normal(size=(n, k))])
        y = X @ np.r_[1.0, beta] + rng.normal(size=n)
```

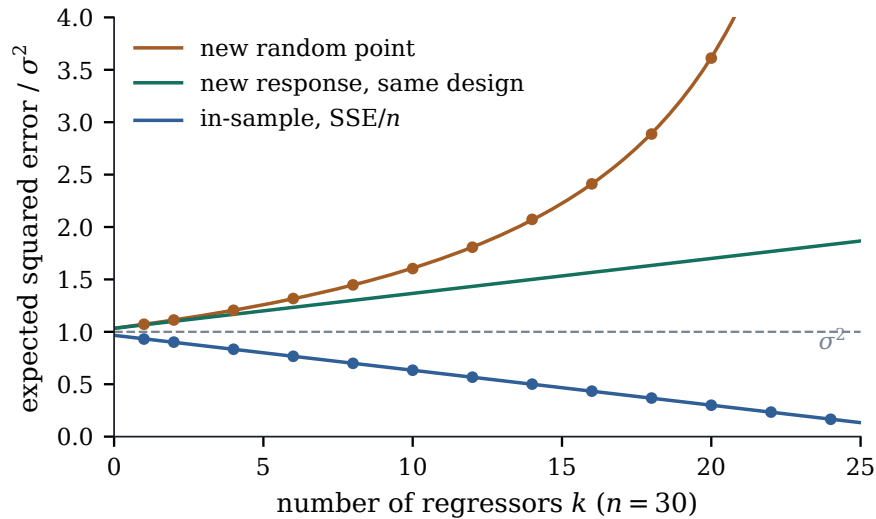


Figure 14.2.1: Expected squared error of a least squares fit, $n = 30$: in-sample (bottom), new responses at the same design (middle) and a new case with normal regressors (top, Eq. 14.2.2). Points: simulation.

```

b, *_ = np.linalg.lstsq(X, y, rcond=None)
ins[s] = np.mean((y - X @ b) ** 2)          # SSE / n
# new point  $X_0 \sim N(0, I)$ ,  $Y_0 = 1 + \beta^T X_0 + e_0$ : exact conditional error
out[s] = sigma2 + (b[0] - 1.0) ** 2 + np.sum((b[1:] - beta) ** 2)
return ins.mean(), out.mean()

results = {}
for k in (1, 5, 10, 20):
    ins, out = simulate(k)
    results[k] = (ins, out)
theory = sigma2 * (1 + 1 / n) * (n - 2) / (n - k - 2)
print(f"k = {k:2d}: in-sample {ins:.3f} (theory {sigma2 * (n - k - 1) / n:.3f}),
      "
      f" new point {out:.3f} (theory {theory:.3f})")

```

[Run this code online](#)

Example. (Predicting doctor visits)

The RAND Health Insurance Experiment records, for 20190 person-years, the number of outpatient visits and characteristics of the insurance plan and of the person. Treat these rows as the population, and predict $\log(1 + \text{visits})$ linearly from 6 of the characteristics (coinsurance rate, deductible and participation-incentive terms, maximum expenditure, physical limitation and a chronic-disease index). The population projection has mean squared error 0.6330 and explains a fraction 0.094 of the variance. Draw training samples of 40 rows, fit by least squares, and evaluate each fit exactly on the whole population. Over 4000 training samples the average in-sample error is 0.5182 and the average error on new cases is 0.8201, against normal-theory values 0.5223 and 0.7705 from Proposition 14.2.7(b). The normal-theory value understates the error on new cases: the regressors are far from normal (two are indicators, one skewed), the linear predictor need not be the conditional mean, and the error variance need not be

constant; the formula assumes none of this. It is a first guide to the optimism, not a substitute for checking it.

```
import statsmodels.api as sm
data = sm.datasets.randhie.load_pandas().data
cols = ["lncoins", "idp", "lpi", "fmde", "physlm", "disea"]
Xpop = np.column_stack([np.ones(len(data)), data[cols].to_numpy()])
ypop = np.log1p(data["mdvis"].to_numpy()) # log(1 + number of visits)
b_pop, *_ = np.linalg.lstsq(Xpop, ypop, rcond=None) # the population projection
sigma2_pop = np.mean((ypop - Xpop @ b_pop) ** 2) # its mean squared error

rng = np.random.default_rng(1423)
n_train, draws = 40, 4000
k_rand = len(cols)
ins, out = np.empty(draws), np.empty(draws)
for s in range(draws):
    idx = rng.integers(0, len(ypop), n_train) # iid draws from the population
    b, *_ = np.linalg.lstsq(Xpop[idx], ypop[idx], rcond=None)
    ins[s] = np.mean((ypop[idx] - Xpop[idx] @ b) ** 2)
    out[s] = np.mean((ypop - Xpop @ b) ** 2) # exact error over the
    population
print(f"population projection error {sigma2_pop:.4f}")
print(f"in-sample {ins.mean():.4f}, new case {out.mean():.4f}")
print(f"normal-theory values: {sigma2_pop * (n_train - k_rand - 1) / n_train:.4f},"
      f" {sigma2_pop * (1 + 1 / n_train) * (n_train - 2) / (n_train - k_rand - 2):.4f}")
```

[Run this code online](#)

14.2.4 Best linear unbiased prediction

So far the new response was independent of the training data. When it is not, as for the next value of a time series or the yield at an unsampled site near sampled ones, the training residuals carry information about the new error, and a good predictor should use it.

Theorem 14.2.8 — Best linear unbiased predictor

Let $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and $y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + e_0$, with $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $E(e_0) = 0$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ for a known positive definite $\mathbf{V}$, $\text{Cov}(\boldsymbol{\varepsilon}, e_0) = \sigma^2 \mathbf{v}_0$ and $\text{Var}(e_0) = \sigma^2 v_{00}$. Suppose $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$. A predictor $\mathbf{a}^\top \mathbf{Y} + a_0$ is **unbiased** if $E(\mathbf{a}^\top \mathbf{Y} + a_0 - y_0) = 0$ for every $\boldsymbol{\beta}$. Among unbiased linear predictors, the mean squared error $E(y_0 - \mathbf{a}^\top \mathbf{Y} - a_0)^2$ is minimized by

$$\hat{y}_0 = \mathbf{x}_0^\top \tilde{\boldsymbol{\beta}} + \mathbf{v}_0^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\tilde{\boldsymbol{\beta}}), \quad (14.2.3)$$

where $\tilde{\boldsymbol{\beta}}$ solves the generalized least squares equations $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \tilde{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$. With $\mathbf{c} = \mathbf{x}_0 - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{v}_0$, its mean squared error is $\sigma^2 [v_{00} - \mathbf{v}_0^\top \mathbf{V}^{-1} \mathbf{v}_0 + \mathbf{c}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{c}]$.

Proof. Put $\delta = \mathbf{V}^{-1}\mathbf{v}_0$ and split the prediction error of any linear predictor:

$$y_0 - \mathbf{a}^\top \mathbf{Y} - a_0 = \underbrace{(e_0 - \delta^\top \boldsymbol{\varepsilon})}_u + \underbrace{(\mathbf{x}_0^\top \boldsymbol{\beta} - \delta^\top \mathbf{X}\boldsymbol{\beta} - (\mathbf{a} - \delta)^\top \mathbf{Y} - a_0)}_w.$$

The term u is uncorrelated with $\boldsymbol{\varepsilon}$, since $\text{Cov}(\boldsymbol{\varepsilon}, e_0 - \delta^\top \boldsymbol{\varepsilon}) = \sigma^2(\mathbf{v}_0 - \mathbf{V}\delta) = \mathbf{0}$. The term w is a constant plus a linear function of $\mathbf{Y}$, hence of $\boldsymbol{\varepsilon}$, so $E(uw) = 0$ and $E(y_0 - \mathbf{a}^\top \mathbf{Y} - a_0)^2 = E u^2 + E w^2$. Only $E w^2$ depends on the predictor.

Unbiasedness means $\mathbf{a}^\top \mathbf{X}\boldsymbol{\beta} + a_0 = \mathbf{x}_0^\top \boldsymbol{\beta}$ for all $\boldsymbol{\beta}$, that is, $a_0 = 0$ and $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$. Equivalently, $(\mathbf{a} - \delta)^\top \mathbf{Y}$ is a linear unbiased estimator of $\mathbf{c}^\top \boldsymbol{\beta}$, which is estimable because $\mathbf{x}_0$ and $\mathbf{X}^\top \delta$ lie in $\mathcal{C}(\mathbf{X}^\top)$. For such predictors $E w = 0$, so $E w^2$ is the variance of $(\mathbf{a} - \delta)^\top \mathbf{Y}$. By Corollary 7.2.3 this variance is smallest when $(\mathbf{a} - \delta)^\top \mathbf{Y} = \mathbf{c}^\top \tilde{\boldsymbol{\beta}}$, the generalized least squares estimator, with variance $\sigma^2 \mathbf{c}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{c}$. The predictor is then $\delta^\top \mathbf{Y} + \mathbf{c}^\top \tilde{\boldsymbol{\beta}} = \mathbf{x}_0^\top \tilde{\boldsymbol{\beta}} + \delta^\top (\mathbf{Y} - \mathbf{X}\tilde{\boldsymbol{\beta}})$, which is Eq. 14.2.3. Finally $E u^2 = \sigma^2(v_{00} - 2\delta^\top \mathbf{v}_0 + \delta^\top \mathbf{V}\delta) = \sigma^2(v_{00} - \mathbf{v}_0^\top \mathbf{V}^{-1} \mathbf{v}_0)$. ■

The predictor is the estimated mean at the new point plus a prediction of the new error e_0 from the residuals, by the best linear predictor of Theorem 2.5.2. In the ordinary model, $\mathbf{V} = \mathbf{I}$ and $\mathbf{v}_0 = \mathbf{0}$, the correction vanishes and Eq. 14.2.3 is the fitted value $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$, with mean squared error $\sigma^2[v_{00} + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0]$. The same theorem, with $\mathbf{v}_0$ built from a spatial covariance function, is the kriging predictor of geostatistics, and with random effects in place of e_0 it gives the predictors of mixed models (Chapter 32, Theorem 32.3.1).

14.2.5 Exercises

14.2.5.1 A. Check your understanding

Exercise A1.

With normal regressors, $n = 50$ and $k = 10$, compute the expected in-sample error $E(\text{SSE}/n)$ and the expected error on a new case, in units of σ^2 . By what factor does the first understate the second?

14.2.5.2 B. Practice

Exercise B1.

For any predictor $g(\mathbf{X})$ with $0 < \text{Var } g < \infty$, let $\tilde{g} = EY + b(g - EY)$ with $b = \text{Cov}(Y, g)/\text{Var } g$. Show that $\tilde{g}$ predicts at least as well as g and that $E(Y - \tilde{g})^2 = \text{Var}(Y)(1 - \text{corr}(Y, g)^2)$. Conclude that among such *linearized* predictors, a higher squared correlation with Y means a smaller mean squared error, and that Eq. 14.2.1 identifies the best of them.

Solution. $\tilde{g}$ is the best linear predictor of Y from the single variable g (Theorem 6.11.1 with $k = 1$), and g itself is one such linear function (intercept 0, slope 1), so $E(Y - \tilde{g})^2 \leq E(Y - g)^2$. By Eq. 6.11.1 the error of $\tilde{g}$ is $\text{Var } Y - \text{Cov}(Y, g)^2/\text{Var } g = \text{Var } Y (1 - \text{corr}(Y, g)^2)$, which is decreasing in the squared correlation. The largest squared corre-

lation is η^2 , attained by $g = m$, whose linearization is m itself.

Exercise B2.

Let the errors of a regression follow a stationary autoregression, $e_t = \phi e_{t-1} + u_t$, $|\phi| < 1$, with white noise u_t , so that $\text{Cov}(e_s, e_t) \propto \phi^{|s-t|}$. Observe $t = 1, \dots, n$ and predict $y_{n+1} = \mathbf{x}_{n+1}^\top \boldsymbol{\beta} + e_{n+1}$. Show that $\mathbf{V}^{-1} \mathbf{v}_0 = \phi \mathbf{e}_n$, the last standard basis vector scaled by ϕ , and hence that the BLUP is $\mathbf{x}_{n+1}^\top \tilde{\boldsymbol{\beta}} + \phi(y_n - \mathbf{x}_n^\top \tilde{\boldsymbol{\beta}})$.

Solution. With $\mathbf{V}_{st} = \phi^{|s-t|}$ (the common factor cancels in $\mathbf{V}^{-1} \mathbf{v}_0$), the covariance of e_s with e_{n+1} is $\phi^{n+1-s} = \phi \cdot \phi^{n-s} = \phi \mathbf{V}_{sn}$. So $\mathbf{v}_0 = \phi \mathbf{V} \mathbf{e}_n$ and $\mathbf{V}^{-1} \mathbf{v}_0 = \phi \mathbf{e}_n$. Substituting into Eq. 14.2.3, the correction term is ϕ times the last residual $y_n - \mathbf{x}_n^\top \tilde{\boldsymbol{\beta}}$. Only the most recent residual matters, because given e_n the earlier errors carry no further linear information about e_{n+1} .

Exercise B3.

In the setting of Corollary 14.2.5, let $s^2 = \text{SSE}/(n - k - 1)$ and let $\mathbf{W}$ be the centred cross-product matrix. Show that $s^2 \mathbf{W}^{-1}$, the usual estimate of the covariance of the slopes, is an unbiased estimator of their *unconditional* covariance $\sigma^2 \boldsymbol{\Sigma}_{XX}^{-1}/(n - k - 2)$.

Solution. Given the regressors, $(n - k - 1)s^2/\sigma^2 \sim \chi^2(n - k - 1)$ (Theorem 7.5.1(b)), a law free of the design, so by Lemma 14.1.1 s^2 is independent of the regressors and hence of $\mathbf{W}$. Therefore $E(s^2 \mathbf{W}^{-1}) = E(s^2) E(\mathbf{W}^{-1}) = \sigma^2 \boldsymbol{\Sigma}_{XX}^{-1}/(n - k - 2)$ by Lemma 14.2.3 and Lemma 14.2.4.

14.2.5.3 C. Going deeper

Exercise C1.

Let the rows $(\mathbf{X}_i^\top, Y_i)$ be jointly normal with k regressors, and fit only the first $k_1 < k$ of them. Show that the expected error on a new case is $\sigma_1^2(1 + 1/n)(n - 2)/(n - k_1 - 2)$, where σ_1^2 is the error variance of the best linear predictor from the first k_1 regressors. *Hint:* under joint normality the smaller model is itself a correct normal model with random regressors. For $n = 30$, $k = 10$, $k_1 = 4$ and $\sigma_1^2 = 1.2\sigma^2$, is the smaller model better for prediction? What is the largest k_1 for which it is?

Solution. The pairs $(\mathbf{X}_{1i}, Y_i)$, with $\mathbf{X}_{1i}$ the first k_1 regressors, are jointly normal, so given $\mathbf{X}_1$ the response follows a normal linear model with k_1 regressors and error variance σ_1^2 (Eq. 3.5.2). The fitted predictor uses only these pairs, so Theorem 14.2.6 applies with k_1 and σ_1^2 in place of k and σ^2 . With the numbers given, the full model has expected error $1.607\sigma^2$ and the smaller one $1.2 \cdot (31/30) \cdot 28/24 \sigma^2 = 1.447\sigma^2$, so the smaller model predicts better despite its larger population error. Since the error of the smaller model increases with k_1 , it wins as long as $1.2 \cdot 28/(28 - k_1) < 28/18$, that is $k_1 < 6.4$: at $k_1 = 6$ its error is $1.578\sigma^2$, at $k_1 = 7$ it is $1.653\sigma^2$. The largest such k_1 is 6.